From one type to another

Hey there! So far, we've learned about typecasting data with SQL as a way of converting data from one type to another in databases. Now I want to check out another way to format data types within spreadsheets. In this video, we'll talk more about why making sure your data is formatted properly is so important and how to format numbers and convert units of measurement in your spreadsheets. Let's get started. Sometimes, you need to convert data when you're working with spreadsheets. That might mean changing numbers into dates, strings, percentages, or even currency.
It's important to double check that all of your data is in the right format for your analysis. Sometimes even after cleaning and processing data, it still might not be in the right format you need. Let's think back to the table with movie data from before. There were a lot of different data types that included numbers, such as dates, budgets, and text strings, like actors' names. These are distinct values, but the spreadsheet doesn't always automatically know that. Here's an example.
Let's say you wanted to sort the movies in this spreadsheet by most recent.
If the spreadsheet cast them as strings instead of dates, it might sort them alphabetically. Until you change the data type, you won't be able to sort them the way you want. It's also possible that your datasets contain inconsistent units of measurement that you'll need to convert. Like say, a table that includes both US dollars and English pounds. That's why it's important to check those data types again, so you don't run into any problems during the actual analysis. Think about the incorrectly cast dates in our movie table. If your boss needed a list of the 20

most recent movies, but your spreadsheet was organized alphabetically instead of by the most recent, you wouldn't be giving her the list of movies she needed. Incorrectly formatted data can lead to time-consuming mistakes in your analysis, and might end up affecting your stakeholders' decision-making. But taking the time early on to convert and format your data can help you avoid that. And now that you know why you'll need to convert data types while working in spreadsheets, let's find out how. First, let me show you a really useful menu for specifying data types in spreadsheets. Here's the movie data table we use before, but now the money columns aren't typed as currency. On the toolbar at the top of the sheet, you'll find a menu that can help you convert these numbers into specific data types. It gives you a lot of choices just from the drop-down menu, such as number, currency, date, percentage....

And if you click to open the full menu, there's even more options, including one for a custom number format. We know that we want these columns to be in currency format, so let's do that. All I have to do is select this column and then hit the currency shortcut. And now it's all typed correctly. But it doesn't stop there. You can go even further and convert the unit of measurement you're using. For this example, let's check out a different table.

Imagine that you're working with a weather channel to gather data about daily temperatures. You have a table with some data about daily observations on the temperature, wind speed, and precipitation in this area. Right now, the temperatures are in Fahrenheit, but for your analysis you need them to be in Celsius. No problem. All you need to do is use the CONVERT function to change the unit of measurement. We'll use this empty column here. Here's the first temperature in the table.

We'll input the CONVERT function in our new column to change it to Celsius. Then we need to put what cell we want converted. And finally, we're going to convert. And presto! Now this cell has the right unit of measurement for your analysis. You can simply apply it to the rest of this column. Now this temperature data is all in Celsius, and your unit of measurement is consistent across the table.

And here's another tip. When adding data to tables using a formula, go back and paste the data in as values afterwards. That way they're locked in. Otherwise the cell stays as a formula and could get confusing when you start working with the data. So let's do that now. We'll copy the values and then right click in a new column. There's an option for "Paste special." And there's an option to "Paste values only."

And now we have the static values in this column. Making sure your data is in the right format before you start analysis is so important. Do this, and your analysis will return the kinds of answers you're really searching for. And now you know some ways to typecast numbers and convert units of measurement in spreadsheets. You can feel confident your data is formatted the right way. Next up, we'll talk more about adjusting your data for analysis and data validation. See you soon.